Answer **all** the questions in the spaces provided.

1 A ball of mass 0.30 kg is moving towards a wall along a straight line at constant speed. It hits the wall and rebounds in the opposite direction. The variation with time t of the momentum p of the ball when it is **not** in contact with the wall is shown in Fig. 1.1

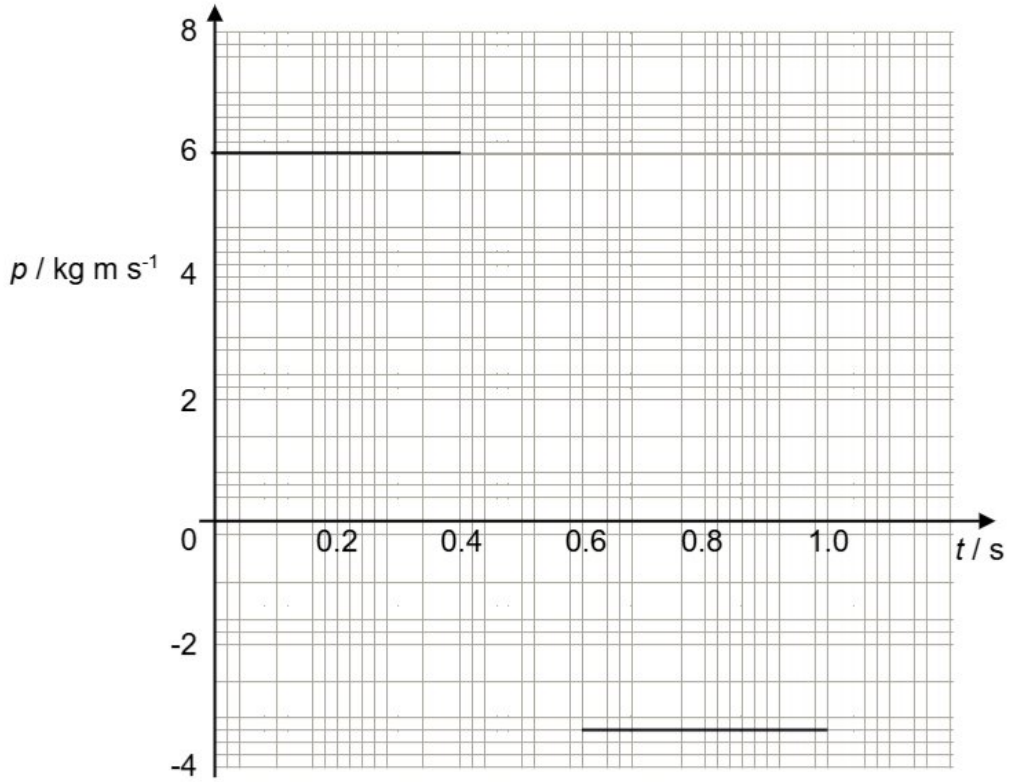


Fig. 1.1

(a) Using Fig 1.1, calculate the impulse experienced by the ball during the collision.

impulse = ... kg m s^{-1} [2]

(b) Calculate the average acceleration of the ball during the collision and state the direction of acceleration with respect to the initial motion of the ball.

acceleration = ... m s^{-2}

direction: ..... [3]

(c) Calculate the change in the kinetic energy of the ball before and after the collision.

change in kinetic energy = ... J [2]

(d) State, with a reason, whether the collision is elastic or inelastic.

.....

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..... [1]

[Total:8]